

>_ Python Operators

A guided path through every operator category in Python – what each one does and the core symbols to remember, in the order you'll typically learn them.

▶ START HERE

+ Arithmetic

Core math on numbers – totals, splits, remainders, powers. Works on **int** and **float**.

`+` `-` `*` `/` `//` `%` `**`

1



Comparison

Compares two values and returns **True/False** – the backbone of every condition.

`==` `!=` `>` `<` `>=` `<=`

2



Assignment

Stores values in variables, and updates them in place – counters, totals, running scores.

`=` `+=` `-=` `*=` `/=` `//=` `%=` `**=`

3



Logical

Combines multiple conditions into one – "both true", "either true", or "flip it".

`and` `or` `not`

4



Membership

Checks if a value exists inside a string, list, tuple, set, or dict – no manual loop needed.

`in` `not in`

5



Identity

Checks if two names point to the **same object** in memory – most often used with **None**.

`is` `is not`

6



Bitwise

Works on raw binary bits – flags, masks, low-level performance, cryptography.

`&` `|` `^` `~` `<<` `>>`

7



String

Builds, searches, cleans, and validates text – the most-used category day to day.

`+` `*` `in` `len()` `.lower()` `.upper()` `.strip()`
`.split()` `.join()` `.replace()` `.find()` `.count()`
`.isdigit()` `.isalpha()` `.isalnum()` `f'{'`

8



Special

Modern shortcuts: assign inside an expression, one-line conditionals, spread collections apart.

`:=` `if / else` `*` `**` `@`

9

✓ YOU KNOW THE OPERATORS



Precedence, highest → lowest

`()` → `**` → `~ +x -x` → `* / // %` → `+ -` → `<< >>` → `&` → `^` → `|` → `comparisons / is / in` → `not` → `and` → `or`